

Hsiu-Hsuan (Jacky) Yeh

Quantitative research · econometrics, machine learning, large-scale data

0jackyyeh@gmail.com | github.com/githubjacky | githubjacky.github.io | linkedin.com/in/hsiu-hsuan-yeh

EDUCATION

National Taiwan University

M.A. in Economics

B.A. in Economics

Taipei, Taiwan

Sep 2023 – Aug 2025

Sep 2019 – Jul 2023

EXPERIENCE

National Taiwan University

Research Assistant

Taipei, Taiwan

Jul 2022 – Jul 2025

Large-scale mobility & panel data engineering

Jul 2023 – Jul 2025

- Cut user-defined-function runtime by up to **90%** over **0.5M+** individuals with Joblib multi-core parallelism and Numba @guvectorize vectorization.
- Built a **50-year** panel of NFL and NBA player, team, and coach records using httpx and selectolax.
- Deployed **rootless Docker with GPU support** on the shared BDSRC server, removing root from every user container.

Stack Python · Polars · PySpark · Joblib · Numba · Docker · Linux

LLM-automated Economic Policy Uncertainty index

Jul 2023 – Jan 2024

- Classified **7,000+** news articles by few-shot LLM prompting through LangChain.
- Fine-tuned OpenAI models on **40,000+** labelled articles.
- Collected metadata for **0.5M+** articles via Playwright automation.
- Published at **CIKM '24**, acceptance rate **26.76%** (141/527).

Stack Python · LangChain · OpenAI fine-tuning · MLflow · Playwright · MongoDB

StochasticFrontiers.jl refactor

Jul 2022 – Jun 2023

- Rebuilt **7,000+ lines** of Julia, decomposing **100+** monolithic functions into self-contained units.
- Introduced multiple dispatch and multithreaded parallelization across the estimation routines.

Stack Julia

Institute of Economics, Academia Sinica

Research Assistant

Taipei, Taiwan

Nov 2022 – Mar 2024

ML treatment assignment for policy evaluation

Nov 2022 – Mar 2024

- Assigned **0.5M+** workers (minimum wage) and **11M+** individuals (estate tax) to treatment and control groups.
- Accelerated XGBoost and Random Forest fits with NVIDIA RAPIDS cuML on GPU.
- Ran **3,000+** hyperparameter-tuning experiments tracked in Optuna and MLflow.

Stack Python · XGBoost · Random Forest · Elastic Net · RAPIDS cuML · Optuna · MLflow

Spatio-temporal Analysis of Telecommunications Data

M.A. Thesis, NTU · 2025

Effects of residential shifts and smartphone adoption on mobility and communication behaviour. Built on **200+ GB** of call detail records covering **300,000+** individuals over **12 months**, with DBSCAN for spatial clustering.

· **Taiwan Economic Association Master's Thesis Award** — Honorable Mention, Econometrics, Nov 2025.

Automation of Text-Based Economic Indicator Construction

CIKM '24 · Short Paper

Yeh, H.-H., Huang, Y.-L., Park, Z., and Chen, C.-C. [A Pilot Exploration on Economic Policy Uncertainty Index](#). *Proc. 33rd ACM Int. Conf. on Information and Knowledge Management*, Boise, ID. Acceptance rate **26.76%** (141/527).

SELECTED PROJECTS

QUANTITATIVE FINANCE

- bitcoin-variance-analysis** Decomposed Bitcoin price and volatility dynamics with Random Forest and GARCH, ranking drivers by Shapley value.
- markov-switching** Identified regime switches in cryptocurrency and equity series with Markov-switching models in Julia.
- fomc-hawkish-dovish** Compared **3** pooling strategies over frozen BERT representations for hawkish/dovish FOMC classification.

MACHINE LEARNING & NUMERICAL METHODS

- course-recommendation** Ensembled ALS, BPR, and KNN collaborative filtering with BM25 retrieval and re-ranking; evaluated by MAP@50.
- BasicIntegration.jl** Julia package covering **4** integration schemes, selecting the method automatically from the domain.
- llm-research** Python package over LangChain exposing a few-shot prompting API with MLflow tracking.

DATA ACQUISITION & TOOLING

- calibrate-google-trends** Pipeline constructing discussion indicators from the unofficial Google Trends API.
- ptt-crawler** Configurable PTT crawler for posts, metadata, push/boo counts, and comments.
- bigkinds-loader** Loader for large-scale Korean news from BigKinds with structured storage.
- ArchSetup** Automated Arch Linux installation scripts and a full Neovim IDE configuration.

SKILLS

- Languages** Python, Julia, R, Stata
- Modelling** XGBoost, Random Forest, Elastic Net, DBSCAN, BERT / PyTorch Lightning, Shapley values, GARCH, Markov-switching, causal inference
- LLM** LangChain, OpenAI fine-tuning, MLflow, Optuna
- Data** Polars, pandas, PySpark, Joblib, Numba, MongoDB, httpx, selectolax, Playwright
- Infrastructure** Git, Docker (rootless, GPU), NVIDIA RAPIDS cuML, Linux (Arch), Neovim